

## SAFETY DATA SHEET

Creation Date 07-May-2010

Revision Date 24-Dec-2021

Revision Number 4

### 1. Identification

**Product Name** Tetrahydro-4 H -pyran-4-ol, 98%

**Cat No. :** AC189430000; AC189430010; AC189430050

**CAS No** 2081-44-9  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                  |            |
|--------------------------------------------------|------------|
| Acute oral toxicity                              | Category 4 |
| Acute Inhalation Toxicity - Vapors               | Category 4 |
| Skin Corrosion/Irritation                        | Category 2 |
| Serious Eye Damage/Eye Irritation                | Category 2 |
| Specific target organ toxicity (single exposure) | Category 3 |
| Target Organs - Respiratory system.              |            |

#### Label Elements

**Signal Word**  
Warning

**Hazard Statements**

Causes skin irritation  
Causes serious eye irritation  
May cause respiratory irritation  
Harmful if swallowed or if inhaled

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing  
Call a POISON CENTER or doctor/physician if you feel unwell

**Skin**

IF ON SKIN: Wash with plenty of soap and water  
If skin irritation occurs: Get medical advice/attention  
Take off contaminated clothing and wash before reuse

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
If eye irritation persists: Get medical advice/attention

**Ingestion**

IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell  
Rinse mouth

**Storage**

Store in a well-ventilated place. Keep container tightly closed  
Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

### 3. Composition/Information on Ingredients

| Component                | CAS No    | Weight % |
|--------------------------|-----------|----------|
| Tetrahydro-4H-pyran-4-ol | 2081-44-9 | 100      |

### 4. First-aid measures

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention immediately if symptoms occur.

**Skin Contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention if symptoms occur.

**Inhalation**

Remove to fresh air. Get medical attention immediately if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if

symptoms occur.

**Most important symptoms and effects**

No information available.

**Notes to Physician**

Treat symptomatically

## 5. Fire-fighting measures

**Suitable Extinguishing Media** Carbon dioxide (CO<sub>2</sub>). Powder. Alcohol resistant foam. Water mist may be used to cool closed containers.

**Unsuitable Extinguishing Media** No information available

**Flash Point** 190 °C / 374 °F

**Method -** No information available

**Autoignition Temperature** No information available

**Explosion Limits**

**Upper** No data available

**Lower** No data available

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

**Specific Hazards Arising from the Chemical**

Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2

**Flammability**  
1

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions** Ensure adequate ventilation. Use personal protective equipment as required. Avoid contact with skin, eyes or clothing.

**Environmental Precautions** Prevent further leakage or spillage if safe to do so. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up** Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

**Handling** Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Wash hands before breaks and immediately after handling the product.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong oxidizing agents. Strong bases.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Engineering Measures** Ensure that eyewash stations and safety showers are close to the workstation location.

#### **Personal Protective Equipment**

**Eye/face Protection** Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection** Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

### **9. Physical and chemical properties**

|                                               |                             |
|-----------------------------------------------|-----------------------------|
| <b>Physical State</b>                         | Liquid                      |
| <b>Appearance</b>                             | Clear, Colourless           |
| <b>Odor</b>                                   | No information available    |
| <b>Odor Threshold</b>                         | No information available    |
| <b>pH</b>                                     | No information available    |
| <b>Melting Point/Range</b>                    | No data available           |
| <b>Boiling Point/Range</b>                    | 87 °C / 188.6 °F @ 15 mm Hg |
| <b>Flash Point</b>                            | 190 °C / 374 °F             |
| <b>Evaporation Rate</b>                       | No information available    |
| <b>Flammability (solid,gas)</b>               | Not applicable              |
| <b>Flammability or explosive limits</b>       |                             |
| <b>Upper</b>                                  | No data available           |
| <b>Lower</b>                                  | No data available           |
| <b>Vapor Pressure</b>                         | No information available    |
| <b>Vapor Density</b>                          | No information available    |
| <b>Specific Gravity</b>                       | 1.070                       |
| <b>Solubility</b>                             | No information available    |
| <b>Partition coefficient; n-octanol/water</b> | No data available           |
| <b>Autoignition Temperature</b>               | No information available    |
| <b>Decomposition Temperature</b>              | No information available    |
| <b>Viscosity</b>                              | No information available    |
| <b>Molecular Formula</b>                      | C5 H10 O2                   |
| <b>Molecular Weight</b>                       | 102.13                      |

### **10. Stability and reactivity**

|                                         |                                                         |
|-----------------------------------------|---------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available              |
| <b>Stability</b>                        | Stable under normal conditions.                         |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat.                     |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Strong bases                   |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ) |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                |
| <b>Hazardous Reactions</b>              | None under normal processing.                           |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

**Toxicologically Synergistic** No information available

#### Products

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------------------|-----------|------------|------------|------------|------------|------------|
| Tetrahydro-4H-pyran-4-ol | 2081-44-9 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

**Persistence and Degradability** Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its volatility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

Not regulated

### TDG

Not regulated

### IATA

Not regulated

### IMDG/IMO

Not regulated

## 15. Regulatory information

### United States of America Inventory

| Component                | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|--------------------------|-----------|------|-----------------------------------------------|-----------------------------|
| Tetrahydro-4H-pyran-4-ol | 2081-44-9 | -    | -                                             | -                           |

#### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA 12(b)** - Notices of Export                      Not applicable

### International Inventories

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                | CAS No    | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL |
|--------------------------|-----------|-----|------|-----------|-------|------|------|------|-------|------|
| Tetrahydro-4H-pyran-4-ol | 2081-44-9 | -   | -    | 218-210-8 | -     | -    | X    | -    | -     | -    |

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

**SARA 313**                                              Not applicable

**SARA 311/312 Hazard Categories**              See section 2 for more information

**CWA (Clean Water Act)**                          Not applicable

**Clean Air Act**                                        Not applicable

**OSHA** - Occupational Safety and Health Administration              Not applicable

**CERCLA**                                              Not applicable

**California Proposition 65**                      This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**              Not applicable

### U.S. Department of Transportation

|                             |   |
|-----------------------------|---|
| Reportable Quantity (RQ):   | N |
| DOT Marine Pollutant        | N |
| DOT Severe Marine Pollutant | N |

**U.S. Department of Homeland Security**              This product does not contain any DHS chemicals.

### Other International Regulations

**Mexico - Grade**                                      No information available

**Authorisation/Restrictions according to EU REACH**

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                | CAS No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Tetrahydro-4H-pyran-4-ol | 2081-44-9 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Tetrahydro-4H-pyran-4-ol | 2081-44-9 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

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## Revision Summary

This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**